XIN BING

PERSONAL INFORMATION

Name: Xin Bing **English first name**: Mike
Email: xin.bing@utoronto.ca
Phone: (607)-697-3443
Current positions: Assistant Professor
Department of Statistics, University of Illinois Urbana-Champaign

Assistant Professor (on leave)
Department of Statistical Sciences, University of Toronto

EDUCATION

Cornell University 09/2016 – 06/2021
Ph.D. in Statistics with minor in Mathematics
Advisors: Florentina Bunea and Marten Wegkamp

University of Washington – Seattle 09/2014 – 06/2016
M.S. in Statistics

Shandong University 09/2013 – 06/2014
M.F.E. in Financial Mathematics

Shandong University 09/2009 – 06/2013
B.S. in Mathematics and Statistics

EMPLOYMENT

Assistant Professor 08/2026 – present
Department of Statistics, University of Illinois Urbana-Champaign.

Assistant Professor (on leave) 06/2022 – present
Department of Statistical Sciences, University of Toronto.

Research Associate 10/2021 – 01/2022
Department of Pure Mathematics and Mathematical Statistics at the University of Cambridge. Hosted by Richard J. Samworth.

Computational Systems Immunology Research Associate 01/2020 – 08/2020
Jointly between School of Medicine at the University of Pittsburgh (hosted by Jishnu Das) and Department of Statistics and Data Science at Cornell University.

RESEARCH OF INTEREST

High-dimensional statistics, mixture models, multivariate analysis, nonconvex optimization, model-based clustering, latent factor model, topic models, minimax estimation, high-dimensional inference, statistical and computational trade-offs, applications of the Wasserstein distance in statistics, applications to genetics, neuroscience, immunology and other areas.

HONORS AND AWARDS

- 2021 Institute of Mathematical Statistics (IMS) Lawrence D. Brown Ph.D. Student Award
- 2021 Hsien and Daisy Yen Wu Award from Cornell University
- Honor graduate of Shandong University in 2013
- Excellent scholarship for meticulous academic performance from 2010 to 2012 at Shandong University

GRANTS

- Connaught New Researcher Award (U of Toronto). ($30K) 2026 – 2028
- Computational and Quantitative Social Sciences Grant from the Data Sciences Institute. ($10K) 2025 – 2026
- Graduate Student Enrichment Scholarship from the Canadian Statistical Sciences Institute (CANSSI). ($15K) 2024 – 2025
- The Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant Individual. ($30K per year) 2023 – 2028
- NSERC Discovery Launch Supplement. ($12.5K) Apr 2023 – Mar 2024

JOURNAL PUBLICATIONS

1. Xin Bing, Dehan Kong and Bingqing Li.
   Convergence and Optimality of the EM Algorithm Under Multi-Component Gaussian Mixture Models.
   Biometrika, 2026+. arXiv link.
2. Xin Bing, Bingqing Li and Marten Wegkamp.
   A New Regression Lens on Multi-Class Classification.
   Bernoulli, 2026+. arXiv link.
3. Xin Bing, Xin He, Dian Jin and Yuqian Zhang.
   Optimal vintage factor analysis with deflation varimax.
   The Annals of Statistics, 53 (4): 1639 - 1666, August 2025. Link to the paper.
4. Xin Bing, Florentina Bunea and Jonathan Niles-Weed.
   Estimation and inference for the Wasserstein distance between mixing measures in topic models.
   Bernoulli, 2025+. arXiv link.
5. Xin Bing and Marten Wegkamp.
   Interpolating Discriminant Functions in High-Dimensional Gaussian Latent Mixtures.
   Biometrika, 111(1): 291-308. March, 2024. Link to the paper.
6. Xin Bing, Wei Cheng, Huijie Feng and Yang Ning.
   Inference in High-dimensional Multivariate Response Regression with Hidden Variables.
   Journal of the American Statistical Association (Theory & Method). Sep, 2023. Link to the paper.
7. Xin Bing and Marten Wegkamp.
   Optimal Discriminant Analysis in High-Dimensional Latent Factor Models.
   The Annals of Statistics 51(3): 1232-1257. June, 2023. Link to the paper.
8. Xin Bing, Florentina Bunea and Marten Wegkamp.
   Detecting approximate replicate components of a high-dimensional random vector with latent structure.
   Bernoulli, 29(2): 1368–1391. May, 2023. Link to the paper.

9. Dian Jin, Xin Bing and Yuqian Zhang.
Unique sparse decomposition of low rank matrices.
IEEE Transactions on Information Theory, 69(4): 2452–2484, April 2023. Link to the paper.

10. Xin Bing, Florentina Bunea, Seth Strimas-Mackey and Marten Wegkamp.
Likelihood estimation of sparse topic distributions in topic models and its applications to Wasserstein document distance calculations.
The Annals of Statistics, 50(6): 3307–3333, December 2022. Link to the paper.

11. Xin Bing, Florentina Bunea and Marten Wegkamp.
Inference in interpretable latent factor regression models.
Bernoulli, 28(2): 997 – 1020, May 2022. Link to the paper.

12. Xin Bing, Yang Ning and Yaosheng Xu.
Adaptive estimation of multivariate regression with hidden variables.
The Annals of Statistics, 50(2): 640–672, 2022. Link to the paper.

13. Xin Bing, Florentina Bunea, Seth Strimas-Mackey and Marten Wegkamp.
Prediction in latent factor regression: Adaptive PCR and beyond.
The Journal of Machine Learning Research, 22(177): 1–50, 2021. Link to the paper.

14. Xin Bing, Florentina Bunea and Marten Wegkamp.
Optimal estimation of sparse topic models.
The Journal of Machine Learning Research, 21(177): 1–45, 2020. Link to the paper.

15. Xin Bing, Florentina Bunea and Marten Wegkamp.
A fast algorithm with minimax optimal guarantees for topic models with an unknown number of topics.
Bernoulli, 26(3): 1765–1796, 2020. Link to the paper.

16. Xin Bing, Florentina Bunea, Yang Ning and Marten Wegkamp.
Adaptive estimation in structured factor models with applications to overlapping clustering.
The Annals of Statistics, 48(4): 2055–2081, 2020. Link to the paper.

17. Xin Bing and Marten Wegkamp.
Adaptive estimation of the rank of the coefficient matrix in high-dimensional multivariate response regression models.
The Annals of Statistics, 47(6): 3157–3184, 2019. Link to the paper.

CONFERENCE PUBLICATION

1. Chao Wang, Xin Bing, Xin He and Caixing Wang.
Towards Theoretical Understanding of Learning Large-scale Dependent Data via Random Features.
International Conference on Machine Learning (ICML), spotlight, 2024. Link to the paper.

2. Dian Jin, Xin Bing and Yuqian Zhang.
Unique sparse decomposition of low rank matrices.
Neural Information Processing Systems (NeurIPS), 2021. Link to the paper.

COLLABORATIVE PUBLICATIONS AND DISCUSSIONS

1. Eugen Pircalabelu and Xin Bing.
Overlapping clustering of time-dependent variables for fMRI data.
J. R. Statist. Soc. C, qlag033, 24 June 2026. Link to the paper.

2. Javad Rahimikollu, Hanxi Xiao, Anna E. Rosengart, Tracy Tabib, Paul Zdinak, Kun He, Xin Bing, Florentina Bunea, Marten Wegkamp, Amanda C. Poholek, Alok V Joglekar, Robert A Lafyatis, Jishnu Das.

SLIDE: Significant Latent Factor Interaction Discovery and Exploration across biological domains. *Nature Methods.* Feb, 2024. Link to the paper.

3. Xin Bing, Tyler Lovelace, Florentina Bunea, Marten Wegkamp, Harinder Singh, Panayiotis Benos, Jishnu Das.
   Essential Regression – a generalizable framework for inferring causal latent factors from multi-omic human datasets.
   Patterns (Cell press), 3(5): 100473, 2022. Link to the paper.

4. Xin Bing, Florentina Bunea, Martin Royer and Jishnu Das.
   Latent model-based clustering for biological discovery.
   iScience 14 (2019), 125–135. Link to the paper.

5. Xin Bing and Marten Wegkamp.
   Discussion of random-projection ensemble classification by Timothy I. Cannings and Richard J. Samworth.
   J. R. Statist. Soc. B 79 (2017), no. 4, 1006–1007.

PAPERS UNDER REVIEW

1. Muheng Li, Peize Zhang and Xin Bing. Nonparametric Maximum Likelihood Estimation for Mixtures of Logistic Regressions. arXiv link.

2. Xin Bing, Florentina Bunea, Marten Wegkamp and Weicheng Wu. Near-Purity Screening and Dominant-Atom Estimation in Softmax Mixtures and Their Ensembles. arXiv link.

3. Xin Bing, Dian Jin and Yuqian Zhang. Federated Independent Component Analysis via Spectral Alignment and Robust Aggregation. arXiv link.

4. Xin Bing and Derek Latremouille. High-Dimensional Invariant Tests of Multivariate Normality Based on Radial Concentration. arXiv link.

5. Xin Bing, Xin He and Chao Wang. Kernel Ridge Regression with Predicted Feature Inputs and Applications to Factor-Based Nonparametric Regression. arXiv link.

6. Xin Bing, Florentina Bunea, Jonathan Niles-Weed and Marten Wegkamp. Learning large softmax mixtures with warm start EM. arXiv link.

WORK IN PROGRESS

1. Softmax-Structured Gaussian Mixtures for Distribution Ensembles. Joint work with Weicheng Wu, Florentina Bunea and Marten Wegkamp. (Preprint available upon request)

2. The Behaviour of High-Dimensional Euclidean Space: Concentration, Geometric Representation, Projection, & Intrinsic Dimension, with Application to Outlier Detection. Joint work with Derek Latremouille. (Preprint available upon request)

3. Iterative Clustering in Low-dimensional Subspace via Multiple Dimension Reductions. Joint work with Dehan Kong and Bingqing Li.

4. Efficient Causality Learning of High-Dimensional Partial Linear Model with Hidden Variables. Joint work with Chao Wang and Xin He.

5. Double denoising k-means clustering. Joint work with Xin He and Shangkai Zhu.

TEACHING AT THE UNIVERSITY OF TORONTO

- STA2211HS: Graduate Probability II 2025-2026, winter

- STA3000Y: Advanced Theory of Statistics 2025-2026, winter
- STA314H: Statistical Methods for Machine Learning I 2025-2026, winter
- STA2211HS: Graduate Probability II 2024-2025, winter
- STA314H: Statistical Methods for Machine Learning I ($\approx$ 360 students) 2024-2025, fall
- STA314H: Statistical Methods for Machine Learning I ($\approx$ 320 students) 2023-2024, fall
- STA314H: Statistical Methods for Machine Learning I ($\approx$ 300 students) 2022-2023, fall

SUPERVISED STUDENTS

Postdoctoral Fellows

- Chao Wang, Aug 2025–present.

Ph.D. Students

- Bingqing Li, Sep 2023–present.
- Muheng Li, Sep 2024–present.
- Weicheng Wu, Dec 2024–present.
- Derek Latremouille, Dec 2024–present.
- Dian Jin (jointly supervised with Yuqian Zhang at Rutgers University; next position: Postdoctoral researcher in the Department of Statistics and Data Science at the National University of Singapore).

Master's and Undergraduate Students

- Peize Zhang (4th year undergrad at the University of Toronto)
- Aditya Aryan (3rd year undergrad at the Indian Statistical Institute)
- Shangkai Zhu (former undergraduate; next position: M.Sc. student at the University of Chicago).
- Yihan Chen (former undergraduate; next position: M.Sc. student at the University of Chicago).

SERVICE

Positions held and service within the University of Toronto

- 2025–2026: Chair of the Organizing Committee for the Department of Statistical Sciences (DoSS) Colloquium.
- 2024–2025: Representative of DoSS at the University-level Connaught Award Meeting.
- 2024–2025: Member of the Organizing Committee for the DoSS Theory and Method Seminar Series.
- 2023–2024: Chair of the Organizing Committee for the DoSS Theory and Method Seminar Series.
- 2023–2024: Member of the Hiring Committee for the tenure-track Assistant Professor position in the Department of Computer and Mathematical Sciences at UTSC.
- 2022–2023: Member of the Organizing Committee for the Statistical Sciences Applied Research and Education Seminar jointly organized by DoSS and CANSSI Ontario.

Positions held and service outside the University of scholarly and academic significance

- 2024–2025: Member of the Organizing Committee for the International Chinese Statistical Association (ICSA) Canada Chapter Symposium 2026 (Montreal, Canada).
- December 2023: Organizer of an invited session at CMStatistics 2023 (Berlin, Germany).

- Referee service: *Annals of Statistics*, *Biometrika*, *Electronic Journal of Statistics*, *Statistica Sinica*, *Journal of the Royal Statistical Society – Series B*, *Journal of the American Statistical Association*, *Journal of Computational and Graphical Statistics*, *Journal of Machine Learning Research*, *SIAM Journal on Imaging Sciences*, and *Journal of Econometrics*.

INVITED TALKS AND PRESENTATIONS

- Institute of Mathematical Statistics (IMS) International Conference on Statistics and Data Science (ICSDS): 1518 December 2026. Split, Croatia. [Invited talk]
- Institute for Mathematical and Statistical Innovation (IMSI) - Workshop: Statistical Foundations of Generative Modeling. 5 – 9, Oct 2026. Chicago, Illinois. [Invited participation]
- The 2026 Joint Statistical Meetings (JSM). 1–6, Aug, 2025. Boston, MA, USA. [Invited talk]
- International Chinese Statistical Association (ICSA): 2026 applied statistics symposium. 14–17 Jun, 2026. Arlington, Virginia. [Invited talk]
- IVADO - Workshop: Statistical Foundations of AI. 8–11 Jun, 2026. Montréal, QC. [Invited talk]
- Seminar in the Department of Statistics and Data Science at the Chinese University of Hong Kong. 05 Feb, 2026. [Departmental seminar talk]
- Colloque des sciences mathématiques du Québec, joint with the seminar talk in the Department of Mathematics and Statistics at McGill University. 09 Sep, 2025. [Invited lecture]
- Seminar in the Department of Statistics and Management at Shanghai University of Finance and Economics. 09 Sep, 2025. Shanghai, China. [Invited talk]
- Seminar in the Institute of Systems Science, Academy of Mathematics and Systems Science at the Chinese Academy of Sciences. 04 Sep, 2025. Beijing, China. [Departmental seminar talk]
- Seminar at the Shandong University Institute for Financial Studies. 29 Aug, 2025. Jinan, Shandong, China. [Departmental seminar talk]
- The 2025 Joint Statistical Meetings (JSM). 2–7, Aug, 2025. Nashville, TN, USA. [Invited talk]
- Seminar in the Department of Statistical Sciences at Cornell University. 22 Apr, 2025. [Departmental seminar talk]
- Seminar in the Department of Statistics and Probability at Michigan State University. 10 Apr, 2025. [Departmental seminar talk]
- Seminar in the Siebel School of Computing and Data Science at the University of Illinois Urbana-Champaign. 7 Apr, 2025. [Departmental seminar talk]
- The 7th International Conference on Econometrics and Statistics: 17-19, July 2024. Beijing Normal University, Beijing, China. [Invited talk]
- The 2nd Joint Conference on Statistics and Data Science in China: 12-14 July, 2024. Haigeng Convention Center, Kunming, Yunnan, China. [Invited talk]
- 2024 Hangzhou International Conference on Frontiers of Data Science: 8-10 July, 2024. Center for Data Science of Zhejiang University, Hangzhou, Zhejiang, China. [Invited talk]
- Oberwolfach Workshop: Statistics and Learning Theory in the Era of Artificial Intelligence. 23–28 June, 2024. Oberwolfach Research Institute for Mathematics, Germany. [Invited talk]
- The 2024 WNAR/IMS/Graybill meeting: 9-12 June, 2024. Colorado State University, Fort Collins, Colorado, U.S.A.. [Invited talk]
- The Sixth ICSA-Canada Chapter Symposium: 7-9 June, 2024. Niagara Falls, Canada. [Invited talk]

- 16th International Conference of the ERCIM WG on Computational and Methodological Statistics (CM-Statistics 2023). HTW Berlin, University of Applied Sciences. Berlin, Germany. 16-18 December, 2023. [Invited talk and organizer of an invited session]
- Seminar in the Department of Mathematics and Statistics at the McMaster University. 31 Oct, 2023. [Departmental seminar talk]
- International Chinese Statistical Association (ICSA): 2023 applied statistics symposium. 11–14 Jun, 2023. Ann Arbor, Michigan. [Invited talk]
- Seminar in the School of Mathematical Sceinces at Shanghai Jiao Tong University. 05 June, 2023. [Departmental seminar talk]
- Seminar in the Department of Mathematics at Shandong University. 22 May, 2023. [Departmental seminar talk]
- Seminar in the Department of Statistics and Management at Shanghai University of Finance and Economics. 28 April, 2023. [Departmental seminar talk]
- Seminar in the Department of Mathematics and Statistics at Auburn University. 19 April, 2023. [Departmental seminar talk]
- Seminar in the Department of Statistics and Actuarial Science at the University of Waterloo. 08 March, 2023. [Departmental seminar talk]
- 15th International Conference of the ERCIM WG on Computational and Methodological Statistics (CM-Statistics 2022). King's College London, UK. 17–19 December 2022. [Invited talk]
- International Chinese Statistical Association (ICSA): 2022 applied statistics symposium. 19–22 Jun, 2022. Gainsville, Florida. [Invited talk]
- Oberwolfach Workshop: Re-thinking High-dimensional Mathematical Statistics. 15–21 May, 2022. Oberwolfach Research Institute for Mathematics, Germany. [Postponed due to pandemic]
- Seminar in the Department of Statistics and Management at Shanghai University of Finance and Economics. 28 April, 2022. [Departmental seminar talk]
- Seminar in the Institute of Data Analysis and Modeling in economics and statistics, UCLouvain. Nov 26, 2021. [Departmental seminar talk]
- Joint Statistical Meetings / IMS Annual Meeting. Seattle, Washington, USA, August 7–12, 2021. [Special talk: IMS Lawrence D. Brown Ph.D. Student Award]
- International Chinese Statistical Association (ICSA): 2020 applied statistics symposium. Houston, TX. [Postponed due to pandemic]
- Workshop at Institute for Advanced Study (IAS, Princeton University): Missing Data Challenges in Computation, Statistics, and Applications. March 16–18, 2020. [Postponed due to pandemic]
- International Chinese Statistical Association (ICSA): 2019 applied statistics symposium. 09 – 12 Jun, 2019. Raleigh, NC. [Invited talk]
- International Center for Mathematical Science (ICMS) workshop: Computational strategies for large-scale statistical data analysis. Date: 02–07 July 2018. Edinburgh, UK. [Invited talk]
- Oberwolfach Workshop: Matrix Estimation Meets Statistical Network Analysis: Extracting low-dimensional structures in high dimension. 17–23 Jun 2018. Oberwolfach Research Institute for Mathematics, Germany. [Contributed talk]